



BI4ALL's Challenges

Ride4All Challenge I



Ride sharing Company

Retail4All Challenge II



Retail Company

Pharma4All Challenge III



Pharmaceutical Company

BI4ALL's Challenges

Ride4All
Challenge I



Ride sharing Company

Retail4All
Challenge II



Retail Company

Pharma4All
Challenge III



Pharmaceutical Company

BI4ALL's Challenges – Challenge II – Retail company



Context: **Retail4all** is a retail company

Retail4all's administration wants to improve its analytical output and improve decision-making processes.

Data sources: **Retail4all's** administration has made available CSV files, which were extracted directly from the main servers of the company. The server is refreshed daily with new data, which allows us to perform updated analysis and faster decision-making. **Retail4all** has about 3 years of historic data.

Challenge: **Retail4all** is looking for solutions that enable the analysis of:

- Sales per month, Quarter and Yearly (accumulated to date)
- Quantity of Sales as well as Amount of Sales
- Comparison of current year vs previous year
- Building client profiles
- Sales details (at the transaction level)
- Aggregate Sales by Store, Category and Subcategory of Products

Security: **Retail4all** needs a solution that implements Row-Level Security of Sales by User/ Location.

BI4ALL's Challenges – Challenge II – Retail company



Data Sources

-  **Stores** (Store ID, Name, Location, Online)
-  **Locations** (Location Code, Location Name, District, Country)
-  **Products** (SKU, Product Name, Category, Subcategory)
-  **VENDAS – Transactions** (Sales)

- ☐ Sale ID
- ☐ Datetime
- ☐ SKU
- ☐ Store
- ☐ POS
- ☐ Operator
- ☐ Currency
- ☐ Qty
- ☐ Amount
- ☐ Dt_Cri

BI4ALL's Challenges – Challenge II – Retail company



Data Sources

Stores (Store ID, Nome, Location, Online)

```
Stores A.csv - Notepad
File Edit Format View Help
"Store ID","Nome","Localização","Online"
"1","Tech-Xchange","1","Y"
"2","Digi-Mart","1","Y"
"3","Silicon Seekers","1","Y"
"4","Gadget Genie","17","Y"
"5","Geek's Haven","17","Y"
"6","Circuit Connection","1","S"
"7","Tech-Tronix","1","N"
"8","Innovation Emporium","1","S"
"9","Digital Digs","25","N"
"10","Geek's Guild","22","N"
"11","Tech-topia","21","N"
"12","Digitopolis","21","N"
"13","Innovationville","1","S"
"14","Circuit City","1","N"
"15","Tech Town","1","N"
"16","Geek-Zone","1","N"
"17","Code Cave","1","N"

Stores B.csv - Notepad
File Edit Format View Help
Store ID,Nome,Localização,Online
21,Farmácia da Alegria,8,N
22,TecnoLoja,8,N
23,Informática Portugal,9,N
24,Eletrónicos Online,9,N
25,Telecomunicações de Portugal,8,S
26,Internet Portugal,4,N
27,Azulejo Tech,4,N
28,Fado Tech,8,
29,Portugal Digital,8,S
20,GastroTech,5,N
31,PortoTech,21,N
```

 Stores

 Locations

 Products

 SALES

 Stores A.csv

25/01/2024 01:00

CSV File

1 KB

 Stores B.csv

25/01/2024 01:00

CSV File

1 KB

BI4ALL's Challenges – Challenge II – Retail company



Detalhe das Fontes de dados

Locations (Location Code, Location Name, District, Country)

```
Locations.csv - Notepad
File Edit Format View Help
1,Aguiar da Beira,,Portugal
2,Alcobaça,,Portugal
3,Albufeira,,Portugal
4,Alcácer do Sal,,Portugal
5,Alcochete,,Portugal
6,Almada,,Portugal
7,Alenquer,,Portugal
8,Alpiarça,,Portugal
9,Alvaiázere,,Portugal
10,Alvito,,Portugal
11,Amarante,,Portugal
12,Anadia,,Portugal
13,Angeja,,Portugal
14,Aveiro,Aveiro,Portugal
15,Avis,,Portugal
```

 Stores

 **Locations**

 Products

 SALES

 Locations.csv

23/01/2024 15:36

CSV File

4 KB

BI4ALL's Challenges – Challenge II – Retail company



Detalhe das Fontes de dados

Products (SKU,Product Name,Category,Subcategory)

```
Products.csv - Notepad
File Edit Format View Help
SKU,Product Name,Category,Subcategory
A12345,Apple iPhone 13 Pro Max,Electronics,Phones
B67890,Samsung Galaxy S23 Ultra,Electronics,Phones
C00111,Sony PlayStation 5,Electronics,Gaming Consoles
D98765,Xbox Series S,Electronics,Gaming Consoles
E43210,Apple iPad Pro 12.9-inch,Electronics,Tablets
F56789,Samsung Galaxy Tab S8 Ultra,Electronics,Tablets
G87654,Sony WH-1000XM5 Headphones,Electronics,Headphones
H32109,Apple AirPods Pro,Electronics,Earphones
I65432,Samsung Galaxy Buds Pro,Electronics,Earphones
J10987,Nike Air Zoom Pegasus 39,Sports & Fitness,Running Shoes
K98765,Adidas Ultraboost 23,Sports & Fitness,Running Shoes
L87654,Under Armour HOVR Machina 3,Sports & Fitness,Running Shoes
M76543,Reebok Floatride Energy 4,Sports & Fitness,Running Shoes
N65432,Nike Dri-FIT Men's Training Crewneck,Clothing & Accessories,Tops
O54321,Adidas Tiro 21 Tracksuit Bottoms,Clothing & Accessories,Bottoms
P43210,Under Armour UA Rival Fleece Joggers,Clothing & Accessories,Bottoms
Q32109,Reebok Classic Leather Sneakers,Clothing & Accessories,Sneakers
```

CSV Stores

CSV Locations

CSV Products

CSV SALES

Products.csv

25/01/2024 01:03

CSV File

7 KB

BI4ALL's Challenges – Challenge II – Retail company



Detalhe das Fontes de dados

SALES – Transactions (Sales)

Rides.csv - Notepad

File Edit Format View Help

```
Ride ID,StartDatetime,EndDateTime,Duration,Car,Driver,Pickup
Location,Dropoff Location,Customer,Currency,Gas,Kms,Amount,Dt_Cri
1,01/01/2020 10:00,01/01/2020
11:10,4200,235,235,72,72,3693,EUR,4.937616105,70.53737293,372.9492918,01/01/
2020 11:11
2,01/01/2020 10:02,01/01/2020
10:06,240,487,487,120,48,6620,EUR,0.310486677,4.43552396,25.27736509,01/01/2
020 10:07
3,01/01/2020 10:03,01/01/2020
11:14,4260,298,298,77,8,4632,EUR,4.98613762,71.23053743,437.8107298,01/01/20
20 11:15
4,01/01/2020 10:05,01/01/2020
11:06,3660,468,468,109,94,8300,EUR,4.327199585,61.81713692,576.5203979,01/01
/2020 11:07
5,01/01/2020 10:07,01/01/2020
```

Ln 1, Col 1 100% Windows (CRLF) UTF-8

CSV Stores

CSV Locations

CSV Products

CSV SALES

Rides.csv

11/01/2024 12:07

CSV File

114,057 KB

Thank you

